MedLift-3D: Click-to-3D Lung Nodule Mass and Growth Assessment from Low-Dose Chest Tomosynthesis with Calibrated Uncertainty

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Abstract

Lung nodules are followed up by measuring how fast they grow, but computed tomography (CT), the standard tool, gives the patient a radiation dose and a cost at every follow-up, while low-dose digital chest tomosynthesis (DTS) cannot resolve depth because its 30° sweep leaves a "missing wedge" of unmeasured information. MedLift-3D is a click-to-3D system that turns one click on a DTS slice into a 3D nodule measurement with a calibrated uncertainty. We first show from projection moments that every DTS view measures a nodule's mass exactly, while its depth extent, which volume needs, is about 15 times worse conditioned; mass and mass doubling time are therefore the measurements DTS can support. The pipeline simulates low-dose 15-view DTS for 583 LIDC-IDRI patients (1,144 radiologist-annotated nodules), reconstructs a 64 mm prism through the chest around the click, fills the missing wedge with a 17.4-million-parameter conditional 3D diffusion model kept consistent with the X-rays, represents the nodule with 3D Gaussians that give its mass in closed form, and wraps every number in conformal prediction intervals and calibrated size-category probabilities. A learned 3D detector suggests where to click, and a seen/unseen (null-space) analysis explains which part of each measurement comes from the X-rays and which from the learned prior. On held-out patients the diffusion prior raised nodule-region PSNR from 11.88 dB to 15.23 dB without overfitting (held-out to training loss ratio 0.97), and the detector reached a CPM of 0.298 against 0.017 for a training-free baseline. **[FILL: one sentence with the H1 result (mass vs volume error) and one with the coverage of the 90% intervals]**. The full study ran on one rented GPU in 14 hours for about US$16, and an interactive demo runs on a free Kaggle notebook.

Keywords: digital tomosynthesis, lung nodule volumetry, diffusion models, 3D Gaussians, conformal prediction, explainable AI

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Chapter 1

Introduction

This chapter states the clinical problem that MedLift-3D solves and why it matters. It then lists the objectives as testable hypotheses, summarises the method and its outcomes, and outlines the rest of the report.

1.1 Project Overview

Lung cancer is the leading cause of cancer death worldwide, with about 2.5 million new cases (12.4% of all cancers) and 1.8 million deaths (18.7% of cancer deaths) in 2022 [1]. Survival depends strongly on the stage at diagnosis: five-year survival is 70–90% for stage I non-small cell lung cancer but only 15–19% for stage IV [2]. Early lung cancer usually appears as a pulmonary nodule, yet most nodules are benign. Radiologists therefore decide by following a nodule over months and measuring how fast it grows, usually as a volume doubling time (VDT) [3, 4].

Computed tomography (CT) is the standard tool for this follow-up, but every scan gives the patient a radiation dose of several millisieverts and costs money and scanner time [5]. Low-dose CT lowers the dose, yet repeated follow-up still adds up: about four radiation-induced cancers per 10,000 screened people, most of it from the follow-up scans [6]. In low-resource settings such as rural Bangladesh, CT is often not available at all.

Digital chest tomosynthesis (DTS) acquires 15–74 low-dose X-ray views during a short tube sweep of about 30°, at roughly 0.1 mSv, on equipment that costs far less than CT [7]. It finds nodules about three times more often than chest radiography [8]. However, a 30° sweep leaves a *missing wedge* of directions that no X-ray passes through (Fig. 1.1), so DTS cannot resolve depth and every 3D size measured from it is partly a guess.

Problem statement. Given a low-dose chest DTS acquisition and one click on a nodule, produce its 3D mass, volume and growth rate together with an uncertainty that is honest, so that a clinician can follow nodules without repeated CT and knows when a measurement cannot be trusted.

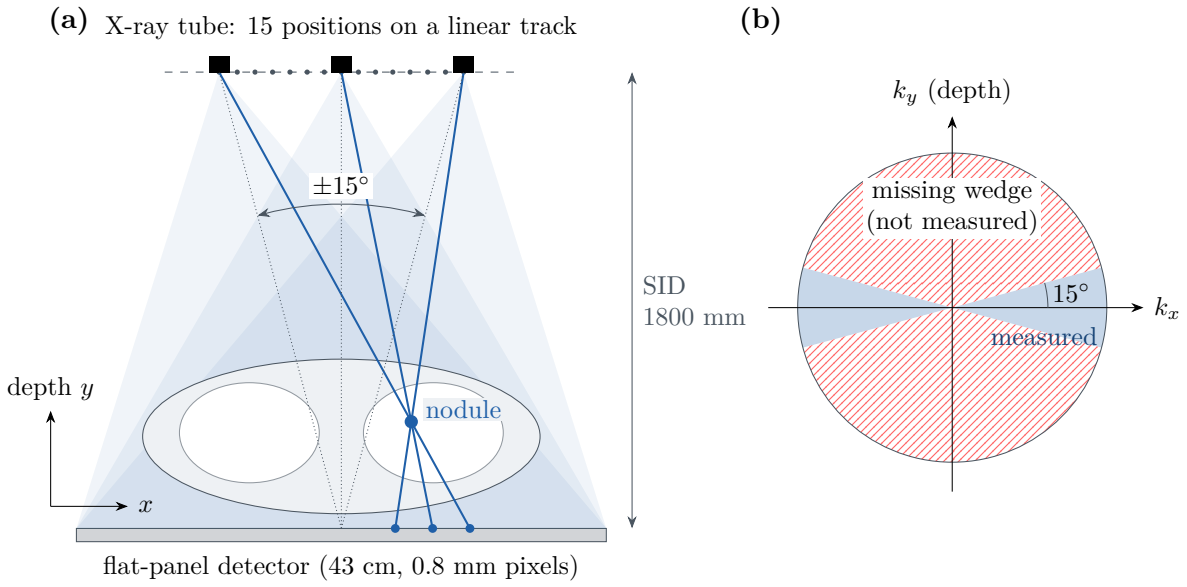


Figure 1.1: Chest DTS acquisition. (a) The X-ray tube moves along a linear track over ±15° and the nodule's shadow shifts on the detector with the tube angle; this shift is the only depth information. (b) In Fourier space each view measures one plane, so a ±15° sweep measures only a double wedge around the in-plane axis and leaves the depth frequencies unmeasured.

1.2 Motivation

Clinical importance. Five-year survival falls from 64% for localised lung cancer to 9% for distant disease [9, 10]. Growth is the key sign separating malignant from benign nodules, and guidelines act on volume and doubling-time categories: < 100, 100–300 and ≥ 300 mm^3, and < 400, 400–600 and > 600 days [4, 11].

Limits of 2D measurement. Diameters measured on single slices assume a sphere. Volumes estimated from 2D measurements differ from 3D volumetry by 47–85% on average [12], 2D measurements of small nodules are unreliable [13], and a 1 mm change in diameter is a 33% change in volume [14]. Measurement error larger than the true growth delays treatment.

Radiation and access. CT follow-up means repeated dose and requires well-equipped centres [5, 15]. DTS is cheaper, faster and much lower in dose, but has no trustworthy 3D measurement.

Why existing reconstruction and AI are not enough. Iterative reconstruction from a 30° sweep smears each nodule 3–5 times longer in depth than in-plane (we measured a median ratio of 3.5 on LIDC nodules). Generative models can fill the missing wedge, but a model that invents anatomy is dangerous for cancer follow-up [16, 17]. Published sparse-view methods report image quality (PSNR, SSIM) rather than calibrated clinical numbers, and none is built for DTS geometry.

Computational motivation. We asked what DTS can *measure*, not only what it can show. A derivation from projection moments (Section 3.2) shows that every DTS view measures a nodule's total excess attenuation, its *mass*, exactly, while its depth extent, which volume needs, is about 15 times worse conditioned for a $\pm 15°$ sweep. A system for DTS should therefore report mass first, fill depth with a learned prior that stays consistent with the X-rays, and state how uncertain each number is. Mass growth is already an established CT biomarker that detects growth earlier than volume [18].

1.3 Objectives

The objectives are stated as testable hypotheses (H1–H6); any of them could turn out false, and Chapter 4 reports what we found.

O1 Mass-first measurement (H1). Show, in theory and on data, that nodule mass from DTS has a smaller error than nodule volume.

O2 Learned prior for the missing wedge (H2). Train a conditional 3D diffusion model that, combined with physics data consistency and 3D Gaussians, reduces volume error compared with physics-only reconstruction.

O3 Calibrated uncertainty (H3, H5). Give every mass and volume a conformal 90% prediction interval that reaches its nominal coverage on held-out patients, and give calibrated probabilities of the clinical size categories with coverage in every size group.

O4 Growth (H4). Measure growth on virtual follow-up scans and test whether mass doubling time (MDT) classifies growth more accurately than VDT.

O5 Explainability (H6). Show for each nodule which part of the 3D result was measured by the X-rays (seen) and which was supplied by the prior (unseen), and test whether the prior's unseen content recovers the truth rather than inventing it.

O6 Usable system. Deliver a click-to-3D application with a learned detector that suggests where to click, running on free or low-cost cloud hardware.

1.4 Methodology

MedLift-3D is evaluated as a virtual clinical trial on real patient CT, because no public dataset has chest DTS together with CT ground truth. Figure 3.5 in Chapter 3 shows the full pipeline.

1. **Data:** 583 LIDC-IDRI patients with 1,144 nodules of 4–30 mm marked by at least three of four radiologists [19, 20], split by patient 60/20/20 into training, calibration and test sets.
2. **DTS simulation:** 15 low-dose views over $\pm 15^\circ$ with Poisson noise, reconstructed by SIRT [21].
3. **Click and prism:** the clinician, helped by a learned detector, selects a nodule; a 64 mm column through the whole chest (the *prism*) is reconstructed around it.
4. **Diffusion prior:** a 17.4-million-parameter conditional 3D U-Net diffusion model [22–24] generates eight CT-like hypotheses of the prism, each kept consistent with the X-rays by SIRT steps inside the sampler.
5. **3D Gaussians:** each hypothesis is fitted with anisotropic 3D Gaussians [25, 26], which give the nodule's mass in closed form.
6. **Uncertainty:** the spread of the eight hypotheses is calibrated into conformal intervals and predictive distributions on the calibration patients [27–30].
7. **Explanation:** each result is split into seen and unseen parts by a conjugate-gradient null-space decomposition [16, 31], and the detector is explained with input×gradient saliency [32].

1.5 Project Outcome

Technical. A verified, open pipeline of 18 scripted steps, each ending in automatic PASS/FAIL checks, from data download to final tables, figures and an HTML report. The full study runs on one rented GPU in about 14 hours for about US$16.

Clinical. A click-to-3D application that returns mass, volume, an EUPS size category with calibrated probabilities and a virtual follow-up, always with intervals instead of single numbers.

Research. (i) A projection-moment argument that DTS measures mass but not depth extent; (ii) a data-consistent diffusion prior inside a DTS prism combined with Gaus-

sian mass; (iii) calibrated, decision-level uncertainty for DTS volumetry; (iv) a truth-referenced seen/unseen test of what the prior adds. Section 2.3 states exactly what is and is not new.

1.6 Organization of the Report

Chapter 2 gives the background on DTS, diffusion models, 3D Gaussians, conformal prediction and explainability, reviews similar applications and research, and identifies the gap. Chapter 3 presents the requirements, the context and data flow diagrams, the user interface, the detailed method with the alternatives considered, the project plan and the task allocation. Chapter 4 describes the environment and tools, how each requirement was verified, and the results for H1–H6. Chapter 5 covers standards; societal, health, safety, legal, cultural, ethical, environmental and sustainability considerations; cost; and the complex engineering problem mapping. Chapter 6 summarises the work, its limitations, future work and what we learned.

Chapter 2

Background

This chapter gives the background needed for the rest of the report and reviews existing applications and research. It ends with the gap that MedLift-3D addresses.

2.1 Preliminaries

X-ray projection and attenuation. A detector pixel records the line integral of the linear attenuation coefficient μ along the ray from the source. With I_0 incident photons the expected count is

$$\bar{I} = I_0 \exp\Big(- \int_{\text{ray}} \mu(\mathbf{r})\, \mathrm{d}l \Big), \qquad I \sim \text{Poisson}(\bar{I}), \tag{2.1}$$

so fewer photons (lower dose) mean more noise. CT values are expressed in Hounsfield units, $\text{HU} = 1000\,(\mu - \mu_{\text{water}})/\mu_{\text{water}}$.

Digital tomosynthesis and the missing wedge. CT collects projections over 180° or more. Chest DTS moves the tube along a short linear track, typically ±15° (GE VolumeRAD) to ±20° (Shimadzu SonialVision), with 15–74 views [7]. By the Fourier slice theorem each view measures one plane of the object's 3D Fourier transform, so a narrow sweep leaves a wedge of depth frequencies unmeasured (Fig. 1.1b). Reconstructions are sharp in the coronal plane but every structure is smeared in depth.

Iterative reconstruction (SIRT). Writing the projection operator as A, the measured line integrals as $\mathbf{b}$ and the volume as $\mathbf{x}$, SIRT [21, 33] solves $A\mathbf{x} \approx \mathbf{b}$ by repeated normalised back-projection of the residual:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \lambda\, C A^{\top} R\Big(\mathbf{b} - A\mathbf{x}^{(k)}\Big), \tag{2.2}$$

where R and C hold the inverse row and column sums of A. It is stable under noise but cannot recover anything that lies in the null space of A.

Seen and unseen parts. Any volume splits uniquely as $\mathbf{x} = \mathbf{x}_{\text{seen}} + \mathbf{x}_{\text{unseen}}$ with $\mathbf{x}_{\text{seen}} \in \mathcal{R}(A^\top)$ and $\mathbf{x}_{\text{unseen}} \in \mathcal{N}(A)$, i.e. $A\mathbf{x}_{\text{unseen}} = 0$. The X-rays determine only the seen part; the unseen part of any reconstruction comes from its prior [16, 31]. Figure 2.1 illustrates this.

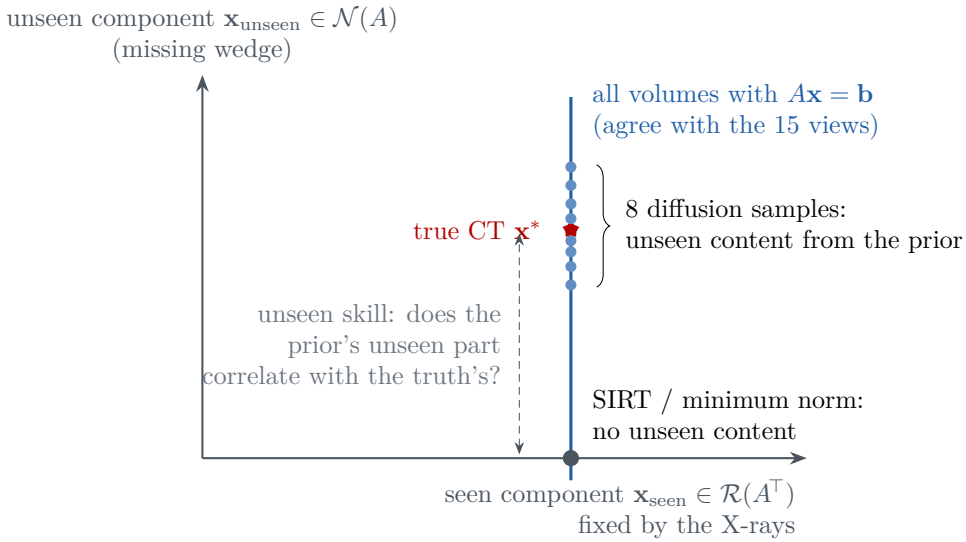


Figure 2.1: Seen (range) and unseen (null-space) components. All volumes on the vertical line agree with the measured views. SIRT returns the minimum-norm point with no unseen content; diffusion samples supply unseen content, which we test against the true CT (unseen skill, Section 3.2).

Denoising diffusion models. A diffusion model learns to reverse a gradual noising process [22]. With a cosine noise schedule $\bar{\alpha}_t$ [34], a noisy sample is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I). \tag{2.3}$$

We train the network to predict the velocity $\mathbf{v} = \sqrt{\bar{\alpha}_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_t}\,\mathbf{x}_0$ (v-prediction [24]), minimising

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}_0,\mathbf{c},t,\boldsymbol{\epsilon}} \left\| \mathbf{v} - \hat{\mathbf{v}}_\theta(\mathbf{x}_t, \mathbf{c}, t) \right\|_2^2, \tag{2.4}$$

where the condition $\mathbf{c}$ is the DTS reconstruction, so the model learns $p(\mathbf{x}_0 \mid \mathbf{c})$. DDIM [23] samples in tens of steps instead of a thousand, and data-consistency (DC) sampling corrects every denoised estimate with the physics so that samples agree with the measurements [35, 36].

3D Gaussian representations. A volume can be written as a sum of anisotropic Gaussians with centres $\boldsymbol{\mu}_i$, covariances $\Sigma_i = R_i S_i S_i^\top R_i^\top$ and amplitudes a_i [25, 26]:

$$f(\mathbf{r}) = \sum_i a_i \exp\Big(-\tfrac{1}{2}(\mathbf{r}-\boldsymbol{\mu}_i)^\top \Sigma_i^{-1}(\mathbf{r}-\boldsymbol{\mu}_i)\Big), \qquad M = \int f = \sum_i a_i (2\pi)^{3/2} s_{i1}s_{i2}s_{i3}, \quad (2.5)$$

where s_{i1}, s_{i2}, s_{i3} are the Gaussian's standard deviations. The mass is therefore closed form, and the field can be evaluated on any grid.

Nodule mass. In this report mass is the nodule's excess attenuation over the surrounding lung, converted to milligrams with the CT density scale (about 0.7–0.85 mg/mm^3 on LIDC nodules). Mass doubling time is computed like VDT but from mass; it detects growth of part-solid nodules earlier [18].

Conformal prediction. Split conformal prediction [27, 30] turns any predictor into intervals with a finite-sample guarantee: if calibration and test cases are exchangeable, a $(1-\alpha)$ interval contains the truth with probability at least $1-\alpha$. A *normalised* version scales each interval by a difficulty estimate (here the spread of the diffusion hypotheses); a *Mondrian* version calibrates within groups (here nodule size groups) so that coverage holds in every group; and *conformal predictive systems* (CPS) return a full calibrated cumulative distribution for each case [28, 29], from which calibrated probabilities such as $P(V \geq 300\ \text{mm}^3)$ follow. Figure 2.2 shows the procedure.

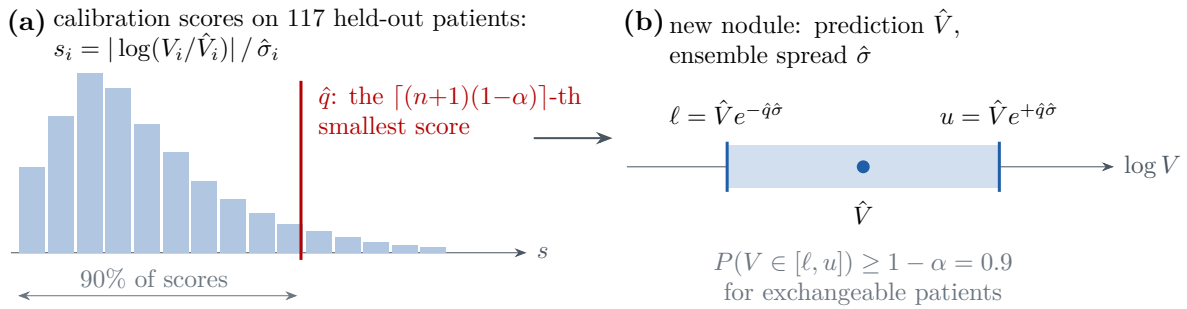


Figure 2.2: Split-conformal calibration (schematic). (a) Normalised scores on held-out calibration patients and their $(1-\alpha)$ quantile $\hat{q}$. (b) The interval for a new nodule, valid for exchangeable patients without any assumption on the error distribution.

Explainability and detection metrics. Input×gradient saliency [32] multiplies each input voxel by the gradient of the output with respect to it, showing which voxels drove a detector's score. Detection is measured by the Competition Performance Metric (CPM), the mean sensitivity at 1/8, 1/4, 1/2, 1, 2, 4 and 8 false positives per scan [37].

2.2 Literature Review

2.2.1 Similar Applications

Table 2.1: Applications similar to MedLift-3D.

Application	What it does	What it lacks for our problem
Chest DTS units (GE VolumeRAD, Shimadzu SonialVision)	Acquire and reconstruct DTS slices for reading	No 3D nodule measurement; depth smear left to the reader
CT nodule volumetry software (vendor lung packages, AI nodule tools)	Detect and segment nodules on CT, report volume and VDT	Requires CT at every follow-up; no DTS support
CT computer-aided detection (LUNA16-style) [37]	Suggest nodule locations on CT	Detection only; no measurement or uncertainty
3D Slicer, ITK-SNAP	Viewing and manual segmentation	No learned prior, no calibrated numbers
Research sparse-view reconstruction codes (R^2-Gaussian, 3DGR-CT) [26, 38]	Reconstruct whole volumes from few views	CT geometry, image metrics only, no clinical numbers

2.2.2 Related Research

Radiation and screening. Mascalchi and Sali [15] reviewed the cumulative dose of low-dose CT screening (9.3–13.0 mSv over ten years) and argued for ALARA and ultra-low-dose methods; Rampinelli et al. [6] estimated the cancer risk from screening and follow-up CT.

3D measurement of nodules. Lee [14] and de Margerie-Mellon et al. [12] showed that volumetry is more reproducible than diameters but varies between software, and Revel et al. [13] showed that 2D measurements of small nodules are unreliable. NELSON [4] and the European position statement [11] define the volume and VDT categories we use. De Hoop et al. [18] showed that mass detects growth of ground-glass nodules earlier than volume or diameter.

Chest DTS. Vikgren et al. [8] found that DTS detected about three times more nodules than chest radiography, and Dobbins and Godfrey [7] described the depth-resolution limit set by the sweep angle. DTS studies measure nodules in 2D or by reader judgement.

Sparse-view CT with deep priors. Score-based and diffusion models have been applied in the sinogram domain [39], with wavelet decompositions [40, 41], for spectral CT [42] and zero-shot with variational solvers [43]; DPS and DDS add data consistency to sampling [35, 36], and foundation models learn lung CT priors from 100,000 scans [44]. Classical iterative reconstruction [33] and compressed sensing [45] suppress noise but over-smooth fine detail, and implicit neural fields [46] are slow per patient. Lighter physics-conditioned diffusion [47], deep image prior with ADMM [48], mixed priors for metal artifacts [49], learned view sampling [50] and unsupervised low-dose denoising [51] address other parts of the problem. All of these report image quality on CT volumes; none produces a calibrated clinical measurement or uses DTS geometry.

3D Gaussians in tomography. 3DGR-CT [38] and R^2-Gaussian [26] reconstruct sparse-view CT with Gaussians, and spatiotemporal Gaussians handle motion [52, 53]. They use Gaussians for speed and quality, not for closed-form mass, and have no learned prior for the missing wedge.

Trustworthiness and uncertainty. Bhadra et al. [16] defined hallucinations through the null space of the imaging operator, DDNM [31] uses a range–null-space decomposition inside diffusion sampling, and Zhang et al. [17] used uncertainty estimates to suppress artifacts in limited-data CT. Conformal predictive systems are available in the open-source crepes library [29] but have not been applied to imaging biomarkers from DTS.

2.3 Gap Analysis

The review shows four gaps. First, DTS research stops at 2D detection and size; no work argues from the projection physics which 3D quantity DTS can measure. Second, generative sparse-view methods target whole CT volumes and image metrics, not a clinical number for one nodule, and none is built for the DTS sweep. Third, Gaussian tomography uses Gaussians for speed, not for closed-form mass. Fourth, reconstruction uncertainty is usually raw variance, which is not calibrated, and explanations of what a prior invented are not checked against the truth. Table 2.3 compares the closest works.

What we do not claim. 3D Gaussians in medical imaging, uncertainty for Gaussian reconstructions, conformal prediction in CT and for lung nodules, null-space hallucination analysis and saliency maps all exist already. Our contribution is their combination for DTS nodule measurement, the mass-first argument, and a calibrated, truth-checked evaluation.

Table 2.2: Summary of the most related research.

Work	Method	Key finding	Gap for DTS nodule measurement
Li et al. [38]	FBP-guided 3D Gaussians	Faster than neural fields	Whole-volume CT; no prior; no measurement
Zha et al. [26]	Radiative Gaussian splatting	Accurate volume integration	No learned prior; no uncertainty
Guan et al. [39]	Sinogram-domain diffusion	Handles varying sparsity	2D; no 3D consistency
He et al. [43]	Zero-shot variational score solver	No paired data needed	Heavy compute; image metrics only
Chung et al. [35, 36]	DPS / DDS sampling	Measurement-consistent samples	Not for DTS; not calibrated
Wang and Guo [44]	Lung CT foundation model	Strong transfer	CT only
Bhadra et al. [16]	Null-space hallucination maps	Formal definition of hallucination	No truth-referenced skill per nodule
Zhang et al. [17]	Progressive artifact learning	Fewer false structures	Uncertainty not calibrated
de Hoop et al. [18]	Nodule mass on CT	Mass detects growth earlier	CT only
Boström [29]	Conformal predictive systems	Valid CDFs for any regressor	Not applied to imaging biomarkers

Table 2.3: Domain comparison: ✓ addressed, ✗ not addressed, ∘ partially.

Work	DTS geometry	Learned prior	Data consistency	Gaussian mass	Clinical biomarker	Calibrated uncertainty	Seen/unseen explanation
Li et al. [38]	✗	✗	✓	✗	✗	✗	✗
Zha et al. [26]	✗	✗	✓	✗	✗	✗	✗
He et al. [43]	✗	✓	✓	✗	✗	✗	✗
Chung et al. [35]	✗	✓	✓	✗	✗	✗	✗
Zhang et al. [17]	✗	✓	✓	✗	✗	∘	✗
Bhadra et al. [16]	✗	✗	–	✗	✗	✗	✓
de Hoop et al. [18]	✗	✗	–	✗	✓	✗	✗
MedLift-3D	✓	✓	✓	✓	✓	✓	✓

2.4 Summary

DTS is low-dose and affordable but blind in depth, and existing AI reconstruction neither respects which quantities DTS can measure nor states how far its numbers can be trusted. MedLift-3D addresses this gap by measuring mass first, filling depth with a data-consistent diffusion prior, calibrating every number with conformal methods, and explaining what was measured and what was supplied by the prior. Chapter 3 turns this into requirements and a design.

Chapter 3

Project Design

This chapter turns the problem into requirements, models the system with context, data flow and decomposition diagrams, and explains the method together with the alternatives we rejected. It ends with the project plan and task allocation.

3.1 Requirement Analysis

Requirements elicitation. Requirements were gathered from four sources: (i) the clinical guidelines that radiologists follow for nodules, namely the EUPS volume categories and the NELSON doubling-time categories [4, 11]; (ii) meetings with our supervisor; (iii) the published limits of DTS and sparse-view reconstruction (Chapter 2); and (iv) the constraints of a student project: public data only, no clinical scanner, a budget of tens of dollars and a fixed deadline. Table 3.1 lists the stakeholders and their needs.

Table 3.1: Stakeholders and their needs.

Stakeholder	Need	How MedLift-3D answers it
Radiologist	A 3D size and growth rate from a low-dose exam, and to know when not to trust it	Mass, volume, category and doubling time, each with a calibrated interval and a low-confidence flag
Patient	Less radiation and cost per follow-up	DTS at about 0.1 mSv instead of CT
Low-resource hospital	Affordable equipment and software	DTS hardware; software runs on a free or US$1/hour GPU
Examiner / researcher	Claims that can be checked	Every step ends in PASS/FAIL checks; test patients never used for tuning

Requirements analysis and negotiation. Several requirements conflict; Table 3.2 records how each conflict was resolved.

Table 3.2: Conflicting requirements and their resolution.

Conflict	Decision	Rationale
Low dose vs accuracy	Simulate $I_0 = 2 \times 10^4$ photons per pixel and view (about 1/20–1/40 of clinical DTS dose)	Conservative: a method that works at this dose works at clinical dose
Clinical view count (60–74) vs compute	15 views at equal total dose	Measured: the view count barely changes depth smear at equal dose (5.13 vs 5.34)
Fill the missing wedge vs never invent anatomy	Diffusion prior with X-ray data consistency, plus seen/unseen explanation	Every hypothesis agrees with the X-rays; what the prior adds is shown and tested
Single number vs honest uncertainty	Eight hypotheses + conformal calibration	Raw spread is not calibrated; conformal intervals have a coverage guarantee
Whole-volume accuracy vs GPU memory	Reconstruct only a 64 mm prism through the chest at the click	Keeps each ray's full path (a small box gave 73% error) at 64×128×64 voxels
Automatic detection vs false positives	The detector only suggests; the clinician clicks	The clinician stays in control; missed nodules can still be measured

3.1.1 Functional and Nonfunctional Requirements

Each requirement has a measurable acceptance criterion and the pipeline step that checks it (Tables 3.3 and 3.4); Chapter 4 reports the result for every row.

Table 3.3: Functional requirements.

ID	Requirement	Acceptance criterion	Checked by
FR-01	Load LIDC-IDRI CT and radiologist annotations; build a consensus mask (≥50% of readers); split by patient into train/cal/test	DICOM and pylidc slice positions agree; splits disjoint	Steps 02–04
FR-02	Simulate low-dose DTS: 15 views, ±15°, SID 1800 mm, 43 cm panel, Poisson noise	Detector integral equals object mass within 0.1% in every view	Steps 01, 05
FR-03	Reconstruct the whole chest with SIRT, confined in depth to the patient	Residual below 0.9 of its start; missing wedge visible (depth/in-plane > 1.3)	Step 06

ID	Requirement	Acceptance criterion	Checked by
FR-04	Suggest nodule candidates with a learned 3D detector	Test CPM higher than a training-free baseline	Step 14
FR-05	Reconstruct a 64 mm prism around a click	Prism keeps the column's attenuation (ratio ≈ 1)	Step 07
FR-06	Generate eight X-ray-consistent CT hypotheses with the diffusion prior	Nodule-box PSNR of samples above the physics-only input on held-out pairs	Steps 08, 15
FR-07	Fit 3D Gaussians; measure mass, volume, 2D diameter and EUPS category	Finite values; failures reported, not hidden	Step 09
FR-08	Calibrate 90% intervals and category probabilities	Test coverage $90 \pm 3\%$; Brier score below the raw ensemble vote	Steps 10, 16
FR-09	Measure growth on virtual follow-ups (VDT, MDT, NELSON categories)	Doubling-time error and category accuracy reported	Step 11
FR-10	Explain each result: seen/unseen split, detector saliency, error drivers	Tables and figures produced on test nodules	Step 17
FR-11	Interactive click-to-3D application and HTML report of all results	Demo self-test passes; report lists every check	Steps 12, 18

3.1.2 Context Diagram

The context diagram (Fig. 3.1) shows MedLift-3D as a single process with its external entities. At run time the clinician is the only actor; the TCIA archive and the cloud GPU provider are needed to build and evaluate the models.

3.1.3 Data Flow Diagram Level 1

The level-1 DFD (Fig. 3.2) decomposes the system into eight processes and four data stores. The upper row runs once to build the cases, train the models and calibrate them; the lower row runs for every click and reads all four stores. Table 3.5 is the data dictionary and Fig. 3.3 the functional decomposition.

Table 3.4: Nonfunctional requirements.

ID	Requirement	Acceptance criterion
NFR-01 Performance	One click gives a result quickly	$\leq$ 60 s per nodule on one GPU
NFR-02 Cost	The full study is affordable for students	$\leq$ US$50 of rented GPU
NFR-03 Reproducibility	Same input, same output; checkable steps	Fixed seed, one configuration file, PASS/FAIL checks, unit tests
NFR-04 Validity	No optimistic bias	Test patients unused for training or tuning; simulation and reconstruction on different grids
NFR-05 Safety	Never report a bare number	Every number has an interval; low-confidence flag when fewer than half of the hypotheses find the nodule or the spread exceeds 10×
NFR-06 Usability	Usable by a clinician without coding	Browser interface, one click per nodule
NFR-07 Portability	Runs on common hardware	CPU smoke test; GPU with $\geq$ 24 GB for training
NFR-08 Privacy	No identifiable patient data	Only de-identified public TCIA data

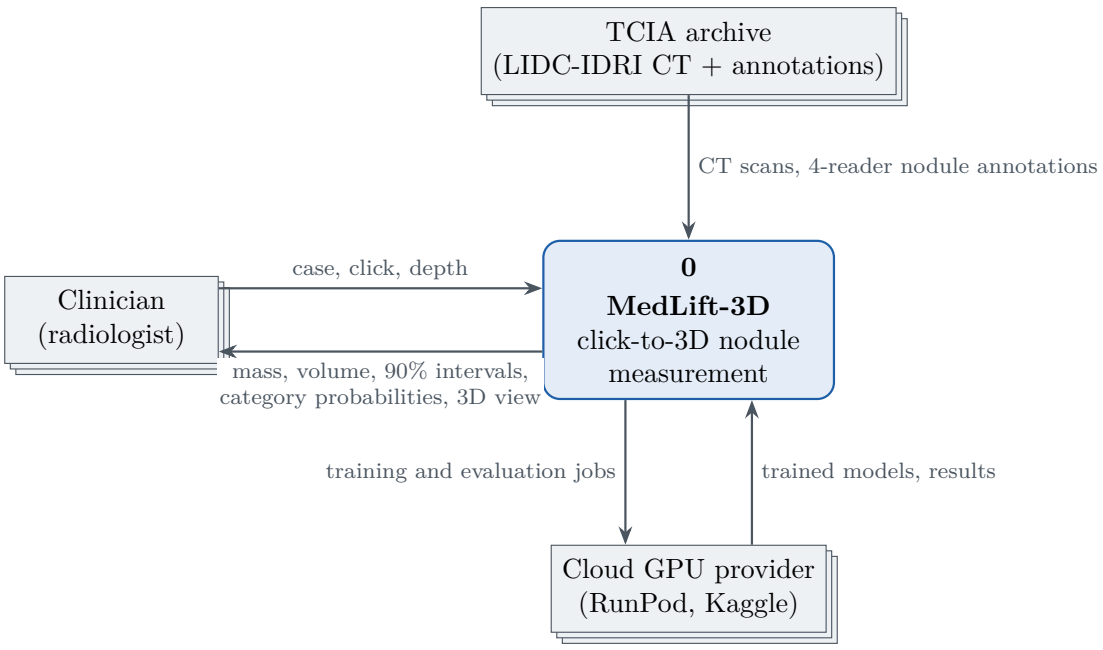


Figure 3.1: Context diagram (DFD level 0) in Gane–Sarson notation.

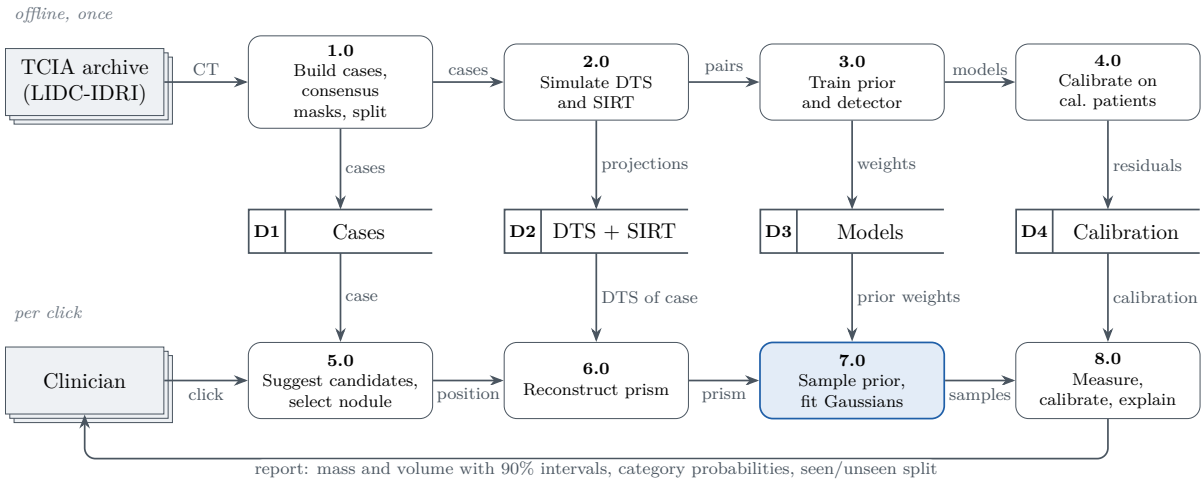


Figure 3.2: Data flow diagram level 1 (offline processes 1.0–4.0, per-click processes 5.0–8.0, data stores D1–D4).

Table 3.5: Data dictionary.

Store / item	Format	Contents	Constraint
D1 Case	int16 + packed bits + JSON header	CT in HU, consensus and four reader masks per nodule, voxel geometry, patient ID	HU in $[-1024, 3071]$
Projections	float, 15×536×536	Noisy line integrals per view	Finite, max < 15
D2 SIRT volume	float, 2.5 mm voxels	Whole-chest DTS reconstruction	Patient depth + 5 mm
Prism	float, 64×128×64, 1×2×1 mm	Column through the chest at the click	Rays $\geq$50% inside
Training pair	float16	Condition (DTS prism) and target (CT prism) in HU	Train patients only
D3 Diffusion model	PyTorch checkpoint	17.42 M parameters, EMA weights	Selected on held-out train pairs
D3 Detector	PyTorch checkpoint	0.34 M parameters, heatmap 3D U-Net	Selected on cal. patients
Gaussians	float32	Centres (N×3), scales (N×3), quaternions (N×4), amplitudes (N)	$N \leq 30{,}000$
Result row	JSON lines	Per nodule and method: volume, mass, log-spread, members, ground truth	One row per nodule
D4 Calibration	JSON	Per method: residuals, spreads, σ floor	Cal. patients only
Report	JSON / UI	Mass and volume with intervals, category probabilities, diameter, low-confidence flag, run time	Interval type stated

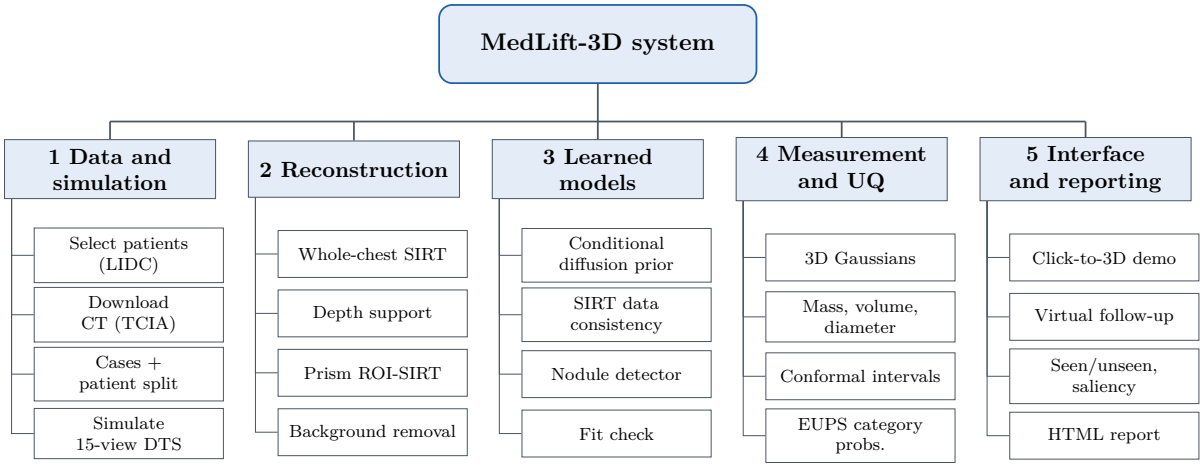


Figure 3.3: Functional decomposition diagram: five subsystems and their functions.

3.1.4 UI Design

The interface is a browser application built around one action: click on the nodule (Fig. 3.4). The numbered parts are: (1) case and method selection; (2) the coronal DTS slice with detector suggestions, annotated nodules (for evaluation) and the click; (3) the depth slider, which supplies the third coordinate; (4) the measurement report with mass and volume intervals, the EUPS category probabilities and the low-confidence flag; (5) a 3D view of the nodule and the prism input with its eight samples; and (6) a virtual follow-up that reports VDT and MDT.

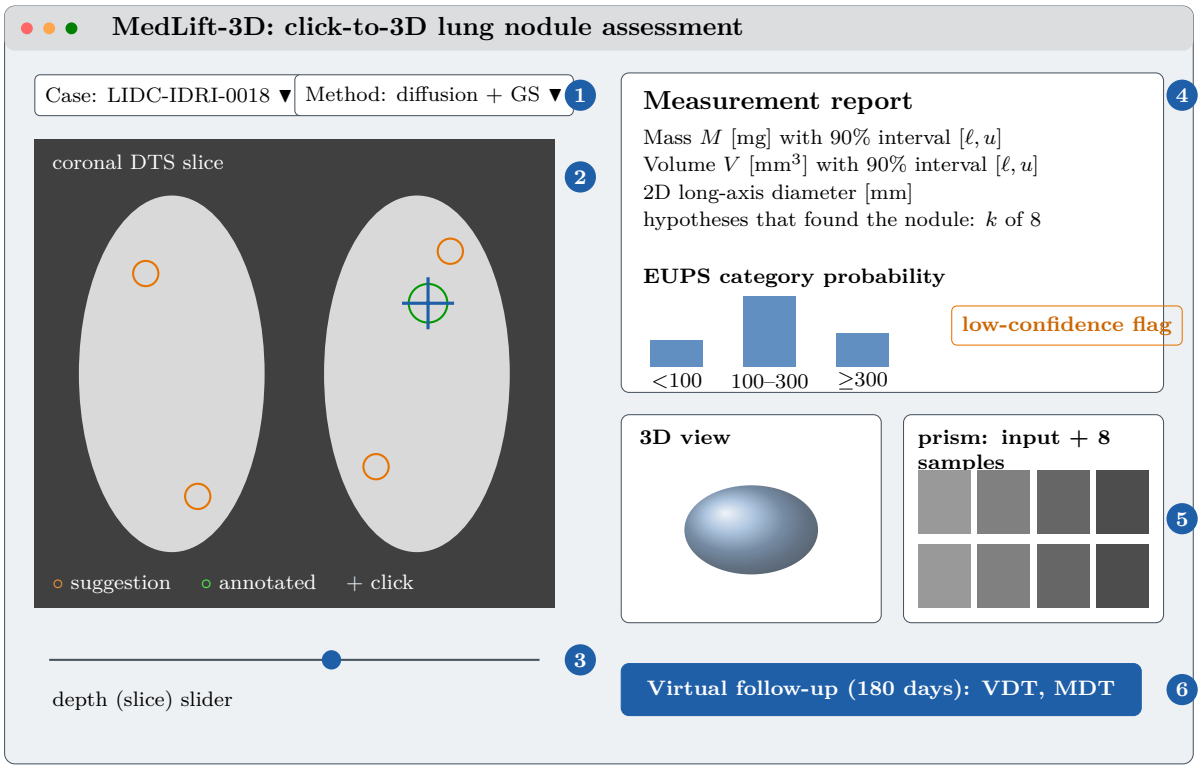


Figure 3.4: User interface design of the click-to-3D application.

3.2 Detailed Methodology and Design

Figure 3.5 gives an overview of the method; the stages are described below.

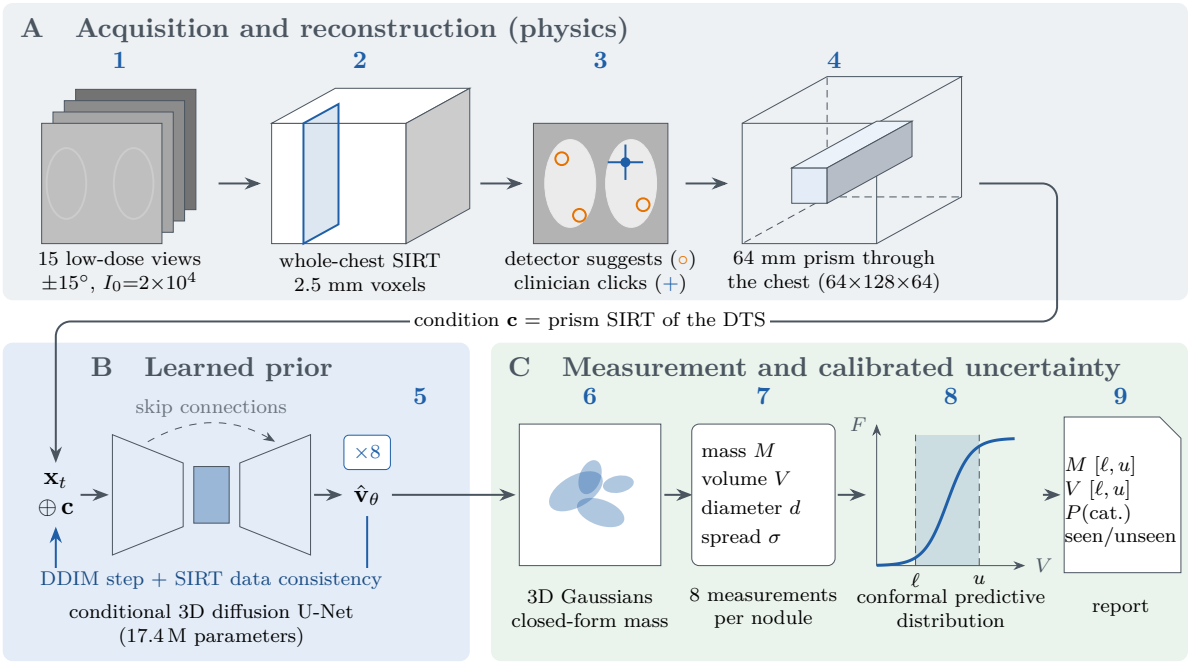


Figure 3.5: MedLift-3D method overview. (A) Physics: low-dose DTS is simulated, reconstructed and cut to a prism around the click. (B) A conditional 3D diffusion prior generates eight hypotheses of the prism, each kept consistent with the X-rays. (C) Gaussians give closed-form mass; the eight measurements are calibrated into intervals and category probabilities.

Stage 0: what DTS can measure. Let $f(x, y, z)$ be the nodule's excess attenuation, with y the depth axis, and let the view at tube angle θ project along a direction tilted by θ from y in the y–z plane, with detector coordinates (x, v) and v along $\mathbf{e}_\theta = (0, -\sin\theta, \cos\theta)$. The moments of the projection p_θ are

$$\iint p_\theta(x, v)\,\mathrm{d}x\,\mathrm{d}v = M, \tag{3.1}$$

$$\iint v\,p_\theta(x, v)\,\mathrm{d}x\,\mathrm{d}v = -\sin\theta\,M y_c + \cos\theta\,M z_c, \tag{3.2}$$

$$\iint (v - \bar{v})^2 p_\theta\,\mathrm{d}x\,\mathrm{d}v = \sin^2\theta\,S_{yy} - 2\sin\theta\cos\theta\,S_{yz} + \cos^2\theta\,S_{zz}, \tag{3.3}$$

where M is the mass, (y_c, z_c) the centroid and S the second central moments (for a cone beam, Eq. (3.1) holds up to a known magnification factor). Every view gives the mass directly (Eq. (3.1)). The depth position enters only through $\sin\theta \le \sin 15°$, making it $1/\sin 15° \approx 3.9$ times worse conditioned than in-plane position, and the depth extent S_{yy}, which volume needs, enters through $\sin^2\theta$, about $1/\sin^2 15° \approx 15$ times worse. Step 01 confirms this numerically: the detector integral matched the mass to 0.03% in all 15 views, and an object three times deeper with equal mass gave the same integrals.

Stage 1: simulated low-dose DTS. The CT table is removed, then 15 views over $\pm 15^\circ$ are simulated at CT resolution with 0.5 mm ray sampling, SID 1800 mm, a 43 cm flat panel with 0.8 mm pixels, monochromatic attenuation and Poisson noise at $I_0 = 2 \times 10^4$. The geometry matches the GE VolumeRAD sweep; reconstruction uses different grids than simulation to avoid the inverse crime.

Stage 2: whole-chest SIRT with depth support. SIRT (40 iterations, 2.5 mm voxels) is confined in depth to the patient (body outline + 5 mm). Without this support the reconstruction placed attenuation in the air in front of and behind the patient, and on real chests the prism lost about a third of its attenuation; with it the prism keeps 98.7%.

Stage 3: learned detector. A small 3D U-Net [54] (0.34 M parameters) predicts a nodule heatmap from the SIRT volume, and its peaks are offered as suggestions. A Laplacian-of-Gaussian and Hessian blob detector is the training-free baseline.

Stage 4: prism ROI. Around the click, a 64×64 mm column through the whole chest ($64 \times 128 \times 64$ voxels at $1 \times 2 \times 1$ mm) is reconstructed by SIRT using only rays that carry at least 50% of their attenuation inside the prism; the rest of the chest is subtracted using the coarse reconstruction. Depth is kept whole because each ray's attenuation is spread along its full path: a small 3D box around the nodule gave a 73% error against 5–7% for the prism.

Stage 5: conditional 3D diffusion prior. A 3D U-Net (Fig. 3.6; base 32 channels, multipliers 1/2/4/4, two residual blocks per level, attention at the lowest level, 17.42 M parameters) learns $p(\text{CT prism} \mid \text{DTS prism})$ with v-prediction, a cosine schedule over 1000 steps, EMA weights and bf16 mixed precision. Sampling uses 50 DDIM steps; from 60% of the way through, every denoised estimate $\hat{\mathbf{x}}_0$ receives two SIRT data-consistency iterations,

$$\hat{\mathbf{x}}_0 \leftarrow \hat{\mathbf{x}}_0 + \lambda C A^\top R\,(\mathbf{b} - A\hat{\mathbf{x}}_0), \tag{3.4}$$

before the next DDIM step. Eight samples give eight hypotheses that agree with the X-rays but differ where the X-rays are blind.

Stage 6: 3D Gaussians and measurement. For each hypothesis the nodule box ($40 \times 50 \times 40$ mm) is fitted with anisotropic Gaussians (100 fitting and 100 refinement iterations against the measured projections, the rest of the prism fixed). A seeded half-maximum segmentation, confined to a plausible nodule extent, gives the volume; Eq. (2.5) gives the mass; and the long-axis 2D diameter is kept as the clinical baseline.

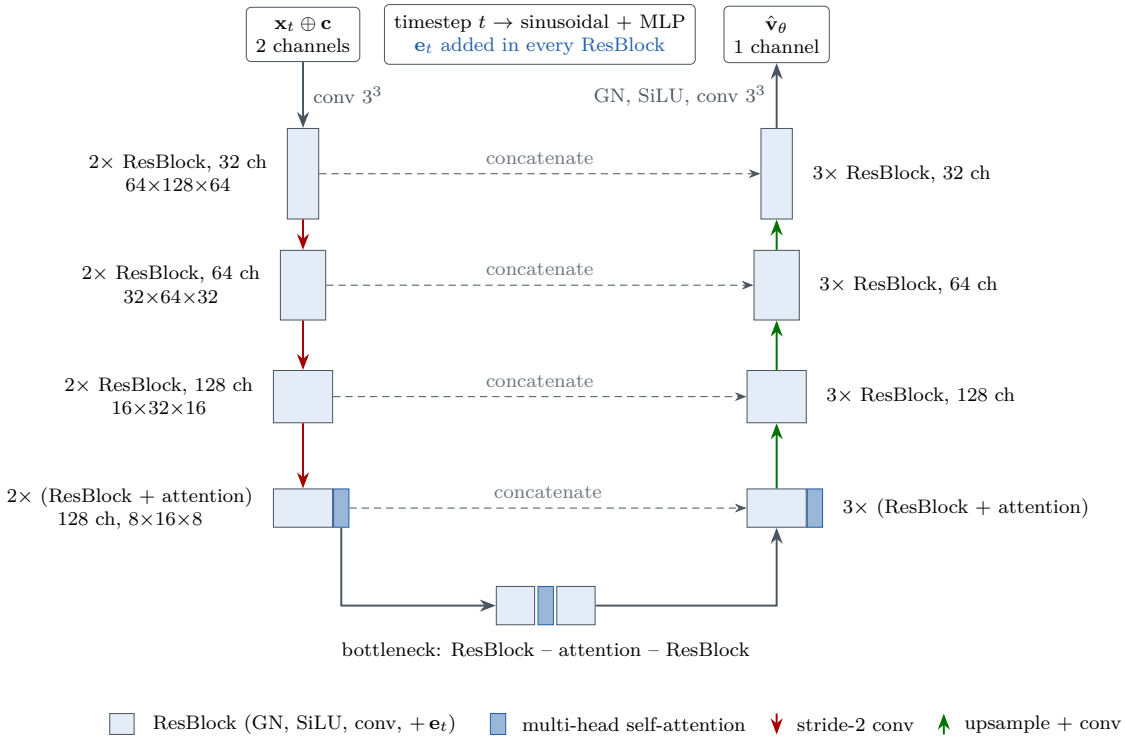


Figure 3.6: Conditional 3D U-Net of the diffusion prior. The noisy prism $\mathbf{x}_t$ is concatenated with the DTS condition $\mathbf{c}$; the timestep embedding is added in every residual block; the network predicts the velocity $\hat{\mathbf{v}}_\theta$.

Stage 7: uncertainty. The eight hypotheses give a log-scale spread $\hat{\sigma}$. On the calibration patients this spread is calibrated with (a) split-conformal asymmetric intervals (step 10) and (b) normalised and Mondrian conformal predictive systems with crepes (step 16), which give full predictive distributions, EUPS category probabilities and category sets.

Stage 8: growth. Each nodule is grown in the CT by a known factor, its projections are updated and fresh noise added; VDT and MDT are measured and assigned to the NELSON categories.

Stage 9: explanation. The true CT prism is split into seen and unseen parts by conjugate-gradient least squares (CGLS) [55] on the prism projection operator. We measure how much of the nodule's mass and shape lies in each part and the prior's *unseen skill*, the correlation of its unseen content with the truth's (Fig. 2.1). Input×gradient saliency explains the detector, and rank correlations relate the measurement error to nodule size, depth and contrast.

Alternatives considered. Table 3.6 lists the alternatives and why they were rejected. The first design of this project, explored in the earlier FYDP phases, generated whole 512^3

volumes with a latent diffusion model; it is included in the table because its lessons shaped the final design: a large generator judged by training losses could not show whether a single nodule measurement was right, so FYDP-III narrowed the goal to clinical numbers that can be verified.

Table 3.6: Alternative solutions and the rationale for the selected design.

Alternative	Why rejected	Selected instead
FBP or SIRT alone	Cannot fill the missing wedge; physics-only volume errors of 80–90% on small nodules	SIRT as input and data-consistency step, plus a learned prior
Neural radiance fields [46]	Slow per-patient optimisation, implicit geometry, no prior for unseen depth	Explicit 3D Gaussians with closed-form mass
Whole-volume latent diffusion at 512^3 (earlier FYDP design)	Too large to train and verify in time and budget; evaluated by generative losses, not clinical numbers	Small conditional prior on a prism, judged by calibrated measurements
Small 3D box ROI	Each ray's attenuation lies mostly outside the box: 73% error vs 5–7%	Prism through the whole chest
GAN prior	Training instability; one output per input	Diffusion: stable training, many hypotheses
Unconditional prior with posterior sampling only	Slower and weaker for a strongly ill-posed 30° problem	Conditional model + SIRT data consistency
Raw ensemble variance or MC dropout	Not calibrated; no coverage guarantee	Ensemble spread calibrated by conformal prediction
Volume-only reporting	Volume depends on the poorly measured depth	Mass first, volume second, both with intervals

3.3 Project Plan

The FYDP-III work was organised in three overlapping streams over the trimester (Fig. 3.7): analysis and design, implementation of the 18-step pipeline, and verification with a full study. Implementation tasks were ordered so that every component could be tested on synthetic phantoms before real data arrived, and real-data pilots on 4 and 40 patients preceded the full 583-patient run.

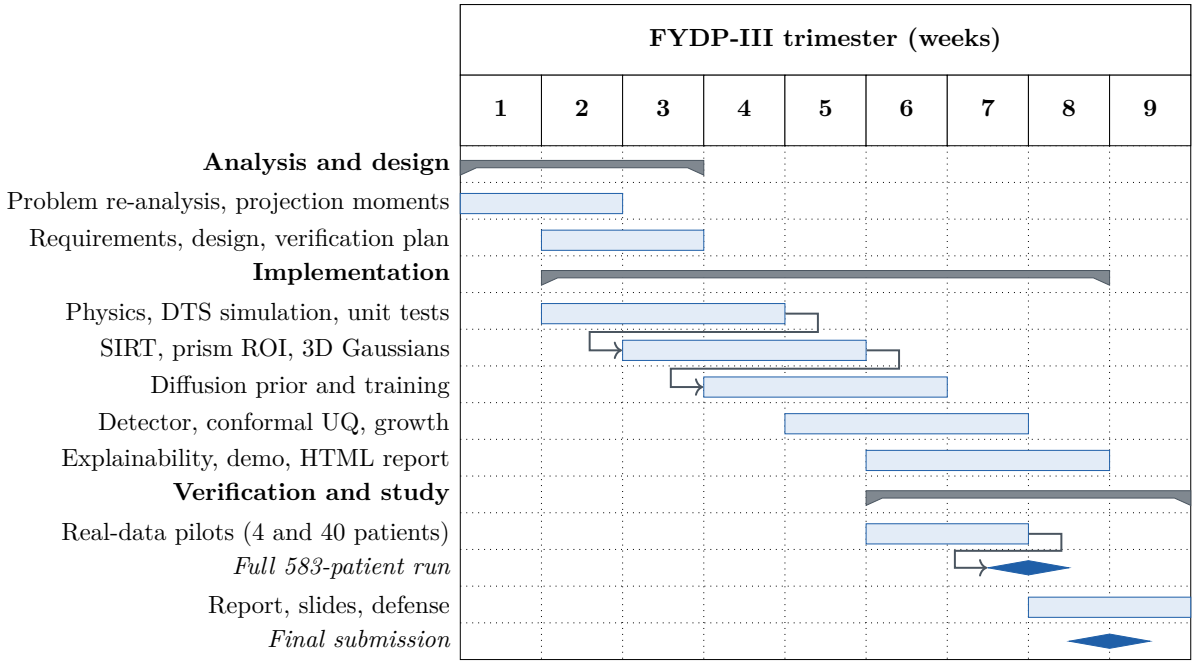


Figure 3.7: FYDP-III schedule by week. **[FILL: adjust the week numbers to your trimester dates]**

3.4 Task Allocation

Table 3.7 gives the responsibilities of each team member in FYDP-III. **[FILL: confirm or correct the allocation]**

Table 3.7: Task allocation matrix.

Team member	Role	Responsibilities
Jubair Ahmed (011222136)	Lead AI architect	Projection-moment analysis; diffusion prior design and training (step 08); fit check (15); conformal uncertainty (10, 16)
Foyez Ahammed Naeem (011222011)	Systems engineer	Cloud infrastructure (RunPod CPU and GPU pods, Kaggle); data pipeline (02–07); detector training (14); full run and backups
Md Motasim Billah (011222016)	Full-stack developer	Click-to-3D application (12); evaluation tables and figures (10, 11); explainability (17); HTML report (18)
All members	–	Literature review, verification of every PASS/FAIL step, report writing, presentation

3.5 Summary

The requirements were elicited from guidelines, the supervisor and published limits, negotiated where they conflicted, and specified with measurable acceptance criteria. The

design measures mass first, reconstructs a prism around the click, fills depth with a data-consistent diffusion prior, measures with Gaussians, and calibrates and explains every number. Chapter 4 verifies each requirement.

Chapter 4

Implementation and Results

This chapter describes the hardware and tools, how every requirement was verified, and the results for hypotheses H1–H6. All numbers come from the full run of October 27, 2026 on held-out test patients unless stated otherwise.

4.1 Environment Setup

Hardware. Table 4.1 lists the computing resources. CPU-only work (downloading about 64 GB of CT and building the cases) ran on a cheap CPU machine, and both machines shared one network volume so that the GPU machine started from the finished cases.

Table 4.1: Computing resources.

Resource	Used for	Specification	Cost
Laptop / Kaggle CPU	Unit tests; synthetic-phantom smoke test	4 cores	Free
Kaggle notebook	40-patient pilot run; final demo and report viewer	2× NVIDIA T4, 16 GB	Free
RunPod CPU pod	Steps 02–04: patient selection, TCIA download, case building	Rented CPU	< US$2
RunPod GPU pod	Steps 05–18: simulation, training, evaluation	NVIDIA L40S, 48 GB	about US$1/h
RunPod network volume	Shared storage between pods; backup after every step	Persistent volume	included

Software and tools. Table 4.2 lists the tools with the alternatives considered.

Configuration and pipeline. One YAML file holds every setting (geometry, dose, prism size, network size, sampling steps, number of hypotheses, $\alpha = 0.1$) and the random

Table 4.2: Software tools and the reasons for choosing them.

Tool	Role	Why chosen (alternative)
Python 3.11+	All code	Standard for medical AI (MATLAB: licence cost)
PyTorch 2.x	Projector, diffusion U-Net, detector, Gaussians; bf16 mixed precision	Differentiable physics and models in one framework
Own ray-driven projector (PyTorch)	DTS forward and back projection with gradients	Exact DTS geometry, differentiable, no CUDA build (ASTRA: separate build, no autograd)
pydicom, pylidc	LIDC DICOM and the four radiologists' annotations	Official annotation access
NumPy, SciPy, scikit-image	Morphology, segmentation, metrics	Mature and tested
crepes, scikit-learn	Conformal predictive systems; calibration metrics	Peer-reviewed implementation [29]
Gradio	Browser application	Quick UI in Python (Flask + JavaScript: more work)
Matplotlib, pytest	Figures; 69 unit tests	Standard
bash, tmux	Unattended 14-hour run with resume and backups	Survives disconnections
Git, GitHub	Version control	Team collaboration

seed is fixed. The pipeline has 18 scripted steps: 00 environment check, 01 physics verification, 02 patient selection, 03 TCIA download, 04 cases and split, 05 DTS simulation, 06 SIRT, 07 training pairs, 08 diffusion training, 09 per-nodule pipeline, 10 evaluation and conformal intervals, 11 growth, 12 demo, 13 geometry sensitivity, 14 detector, 15 fit check, 16 conformal predictive systems, 17 explainability and 18 HTML report. Each step prints PASS/FAIL lines with the expected value.

4.2 Testing and Evaluation

Verification strategy. Testing was done at four levels:

1. **Unit tests (69):** projector adjointness and mass conservation, the Gaussian mass formula, conformal coverage on synthetic data, Mondrian group coverage, CPS probabilities against crepes, the CGLS seen/unseen split, and fit-check parsing.
2. **Smoke test:** the full pipeline on synthetic chest phantoms on a CPU in 10–20 minutes.
3. **Real-data pilots:** 4 patients on CPU and 40 patients on Kaggle, which exposed and fixed the depth-support bug (prism attenuation 67–81% $\rightarrow$ 98.7%).

4. **Full run:** 583 patients with automatic checks after every step and an HTML report listing all checks.

Data split. Patients, not nodules, were split 60/20/20: 350 training, 117 calibration and 116 test patients. The diffusion model was selected on 10% of the training patients held out, the detector on the calibration patients, and the conformal methods were calibrated on the calibration patients. Test patients were used only for the final tables.

Metrics and baselines. Volume and mass error (median signed error, median absolute percentage error MdAPE, Bland–Altman limits), failure rate, nodule-box PSNR, interval coverage and median width, Brier score and expected calibration error (ECE) of category probabilities, doubling-time error and NELSON category accuracy, detection CPM and sensitivity, seen fraction and unseen skill, and run time. Baselines: 2D diameter converted to a sphere volume (clinical practice), coarse SIRT, prism ROI-SIRT, physics-only Gaussians, and diffusion without Gaussians.

Requirement verification. Table 4.3 traces every requirement of Tables 3.3 and 3.4 to its measured result.

Table 4.3: Requirement verification (traceability).

Req.	Criterion	Result	Met
FR-01	Positions agree; splits disjoint	DICOM vs pylidc slice positions 0.000 mm; 350/117/116 patients, disjoint	Yes
FR-02	Integral = mass within 0.1%	0.03% in all 15 views; deeper object of equal mass indistinguishable	Yes
FR-03	Residual < 0.9; depth/in-plane > 1.3	Residual 0.02–0.03 of start; depth smear 2–3.5×	Yes
FR-04	Learned CPM > baseline	0.298 vs 0.017	Yes
FR-05	Prism keeps column attenuation	98.7% after the depth-support fix	Yes
FR-06	Samples beat physics input	Nodule-box PSNR 15.23 vs 11.88 dB (+3.35 dB)	Yes
FR-07	Finite values; failures reported	**[FILL: failure rate per method from table_accuracy]**	**[FILL:]**
FR-08	Coverage $90 \pm 3\%$; Brier $<$ vote	**[FILL: from table_uq_intervals and table_uq_risk]**	**[FILL:]**

Req.	Criterion	Result	Met
FR-09	Growth metrics reported	**[FILL: from table_growth]**	**[FILL:]**
FR-10	Explanation outputs	Seen/unseen, saliency and error-driver tables on 30 test nodules	Yes
FR-11	Demo self-test; report	Demo served from Kaggle; HTML report with 13 checked steps	Yes
NFR-01	$\leq$ 60 s per nodule	40–50 s per nodule for all five methods together on the L40S	Yes
NFR-02	$\leq$ US$50	about US$16 for the full run	Yes
NFR-03	Reproducible, checkable	Fixed seed, one configuration, 69 tests, PASS/FAIL logs	Yes
NFR-04	No optimistic bias	Test patients unused until final tables; separate simulation and reconstruction grids	Yes
NFR-05	No bare numbers	Every value has an interval and a low-confidence flag	Yes

Issues found and fixed. (1) Without a depth support, SIRT placed attenuation in the air around the patient and the prism lost a third of its attenuation as false background; fixed by confining SIRT to the patient. (2) The training-free detector mostly found end-on vessels (recall in the top 10 of 6% in the pilot); replaced by the learned detector. (3) A first CPS implementation capped probabilities at $n/(n+1)$; fixed with the smoothed mid-point p-value and verified against crepes. (4) In step 10 the asymmetric split-conformal upper bound for some methods became effectively unbounded because calibration nodules where a method found nothing (measured volume ≈ 0) produce unbounded log-ratios. **[FILL: state the resolution: failed measurements excluded from calibration and reported as failures, or step-16 CPS intervals used instead]** (5) The HTML report lists three failed checks. **[FILL: name and explain the three failed checks]**

4.3 Results and Discussion

4.3.1 Dataset

4.3.2 Diffusion Prior Training and Fit

Training ran for 27,610 steps within its 6-hour budget (batch 8, about 0.77 s per step). The loss fell from 0.1186 to 0.0476 (mean over the last 10% vs the first 10% of steps).

Table 4.4: Dataset characteristics.

Item	Value
LIDC-IDRI patients (≥1 nodule of 4–30 mm marked by ≥3 readers, slices ≤2.5 mm)	583
Nodules	1,144
Train / calibration / test patients	350 / 117 / 116
Test nodules (detector evaluation)	225 in 116 scans
Nodules measured in step 09	**[FILL: cal / test]**
Consensus volume / mean reader volume	0.81
Single-view nodule contrast-to-noise ratio (median)	3.3

On held-out training pairs the prior's samples reached 15.23 dB PSNR in the nodule box against 11.88 dB for the physics-only input (+3.35 dB), and 13.74 vs 10.16 dB over the whole prism including the chest walls. At fixed noise levels the denoising loss was 0.0340 on training pairs and 0.0329 on held-out pairs, a ratio of 0.97, so the model did not overfit (Fig. 4.1). Its held-out loss was still falling at the end of the budget, so longer training would probably help.

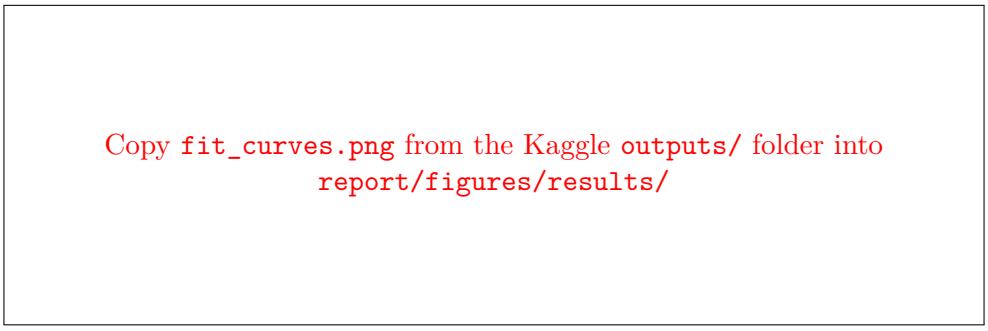


Figure 4.1: Denoising loss at fixed noise levels on training and held-out pairs during training (step 15).

4.3.3 Nodule Detection

Table 4.5: Nodule detection on the 116 test patients (225 nodules).

Detector	CPM	Sensitivity at 1 / 2 / 4 / 8 FP per scan	Recall in top 10
Learned 3D U-Net (ours)	**0.298**	28% / 35% / 40% / 46%	**48%**
Training-free LoG + Hessian	0.017	0% / 1% / 3% / 7%	8%
Learned, in-plane position only (diagnostic)	0.300	28% / 35% / 40% / 47%	49%

The learned detector is 17 times better than the training-free baseline by CPM (Table 4.5). The in-plane-only row is almost identical, so the misses are not suggestions at the wrong depth but nodules that a DTS view does not separate from vessels and chest wall. A sensitivity of 35% at two false positives per scan is far below CT detectors, which is why the detector only suggests and the clinician can click anywhere.

Copy `froc_test.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.2: FROC curves of the learned and training-free detectors on the test patients (step 14).

4.3.4 Measurement Accuracy (H1, H2)

Table 4.6: Measurement error on test nodules against the radiologist consensus. **[FILL: copy from outputs/10_evaluation/table_accuracy.md]**

Method	Volume MdAPE	Mass MdAPE	Failure rate
2D diameter → sphere	**[FILL:]**	–	**[FILL:]**
Coarse SIRT	**[FILL:]**	**[FILL:]**	**[FILL:]**
Prism ROI-SIRT	**[FILL:]**	**[FILL:]**	**[FILL:]**
Physics-only Gaussians	**[FILL:]**	**[FILL:]**	**[FILL:]**
Diffusion	**[FILL:]**	**[FILL:]**	**[FILL:]**
Diffusion + Gaussians (ours)	**[FILL:]**	**[FILL:]**	**[FILL:]**

H1 (mass error < volume error): **[FILL: supported or not, with numbers]**. **H2** (the prior reduces volume error compared with physics-only reconstruction): **[FILL: supported or not, with numbers]**. **[FILL: discuss small vs large nodules and the Bland–Altman bias]**

Copy `volume_errors.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.3: Volume error of each method against the radiologist consensus on test nodules (step 10).

4.3.5 Calibrated Uncertainty (H3, H5)

[FILL: copy outputs/16_uncertainty/table_uq_intervals.md, table_uq_groups.md and table_uq_risk.md: test coverage of the 90% intervals for mass and

Copy `bland_altman_volume.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.4: Bland–Altman plot of volume for the best method (step 10).

volume (target 87–93%), median width, coverage per size group with and without Mondrian, Brier score and ECE of the EUPS probabilities against the raw ensemble vote]

H3: **[FILL:]**. **H5**: **[FILL:]**.

Copy `volume_uq_overview.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.5: Uncertainty overview for volume: prediction intervals, regression with uncertainty, residuals and interval widths (step 16).

Copy `reliability.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.6: Reliability diagram of the calibrated EUPS category probabilities against the raw ensemble vote (step 16).

4.3.6 Growth (H4)

[FILL: copy outputs/11_growth/table_growth.md: VDT and MDT errors and NELSON category accuracy]

H4 (MDT classifies growth better than VDT): **[FILL:]**.

Copy `doubling_time.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.7: Measured vs true doubling time on virtual follow-up scans (step 11).

4.3.7 Explainability (H6)

[FILL: copy outputs/17_explain/table_explain.md and table_error_drivers.md: fraction of nodule mass in the seen part, fraction of shape in the unseen part, unseen skill of the prior with 95% CI, Spearman correlations of error with size, depth and contrast]

For the detector, the share of input×gradient saliency within 10 mm of the suggestion was 0.15 for true positives and 0.16 for false positives: the detector looks at the local region in both cases, and false positives are caused by local structures that resemble nodules (vessels seen end-on), not by attention far from the candidate.

H6: **[FILL:]**.

Copy `seen_unseen_example.png` from the Kaggle `outputs/` folder into `report/figures/results/`

Figure 4.8: Seen/unseen split of the true CT and of the prior's samples for one test nodule (step 17).

4.3.8 Acquisition Geometry Sensitivity

On 12 nodules of two test patients, the depth smear (depth over in-plane width of the artifact spread) was 5.13 for 15 views over 30°, 5.34 for 60 views over 30° (four times the compute), 4.69 for 15 views over 40° and 4.61 for 74 views over 40°, against 0.95 for CT. At equal dose the number of views hardly matters, while a 40° sweep reduces the smear by about 9%; our 15-view, 30° setting is therefore the conservative case.

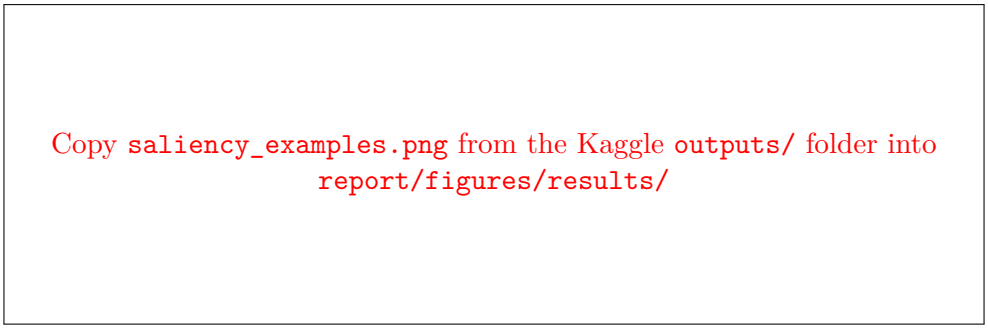


Figure 4.9: Input×gradient saliency of the detector for true and false positive suggestions (step 17).

4.3.9 Run Time, Cost and Application

The full run took 14 hours on one L40S (01:27–15:27, October 27) for about US$16, including 6 hours of diffusion training and about 18 minutes for the detector. The application loads the trained models on a free Kaggle notebook and is reached from a browser through an HTTPS tunnel. On patient LIDC-IDRI-0018 a click on an annotated nodule returned mass and volume with intervals, the EUPS category with calibrated probabilities and the 3D view (Fig. 4.10); a click where no nodule exists returned "no nodule measured at the click" instead of a number, as NFR-05 requires.

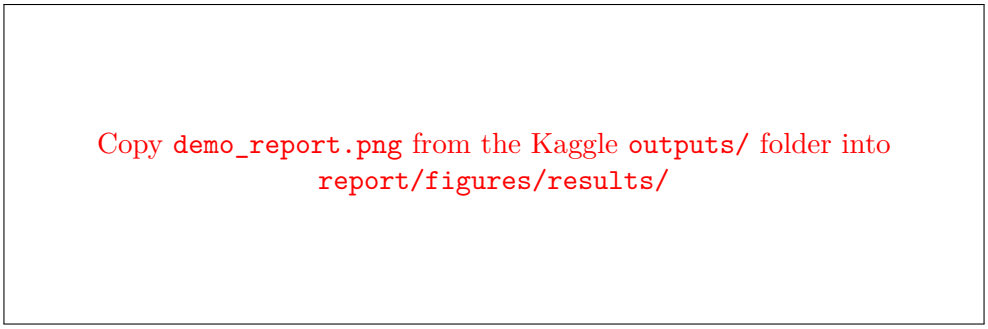


Figure 4.10: The click-to-3D application on a test patient: report with intervals and category probabilities.

4.4 Summary

Every requirement with a fixed criterion was verified by an automatic check. The diffusion prior beat the physics-only input by 3.35 dB in the nodule region without overfitting, the learned detector beat the training-free baseline by a factor of 17 in CPM, and the full study cost about US$16. **[FILL: one sentence each on the H1–H6 outcomes]**. Results that did not support a hypothesis are reported as such.

Chapter 5

Standards and Design Constraints

This chapter lists the standards MedLift-3D follows, with the alternatives considered, and analyses its economic, environmental, ethical, health and safety, social, political, legal, cultural and sustainability aspects. It ends with the cost analysis and the complex engineering problem mapping.

5.1 Compliance with the Standards

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

5.2.1 Economic Constraint

The project had no external funding, so all computation had to fit a student budget. Free resources were used wherever possible (laptop CPU, Kaggle GPUs for the pilot and the application), CPU-only steps (3–5 hours of download and case building) were moved to a cheap CPU machine, and GPU training was given a fixed 6-hour budget. The full study cost about US$16 on the GPU and under US$2 on the CPU machine. Clinically, the economic case rests on DTS: a DTS-capable radiography room costs a fraction of a CT scanner and examines patients faster, so follow-up by DTS can lower the cost of each follow-up for hospitals and patients.

Table 5.1: Software standards.

Standard	How it is applied	Alternatives (pros / cons)	Rationale
DICOM [56]	Input CT read from DICOM; geometry checked against pylidc (0.000 mm difference)	NIfTI: simpler but loses scanner metadata	Hospitals and PACS store images in DICOM
Compressed NumPy with JSON headers	Cases, pairs and results keep their geometry in a header	NIfTI cannot hold per-nodule reader masks with metadata; HDF5 is a heavier dependency	One self-describing file per case
IEC 62304 [57] (software life cycle)	Not certified; practices followed: requirements traced to tests (Table 4.3), version control, unit tests, documented anomalies	Ad-hoc research code: faster but unverifiable	Needed for any clinical use; tracing now costs little
ISO 14971 [58] (risk management)	Hazard analysis with mitigations (Table 5.4)	No risk analysis	Required by regulators
ISO 13485 [59] (quality management)	Identified as required for a product; out of scope here	–	Applies to a device manufacturer
PEP 8, pytest	Code style; 69 automated tests	No tests: faster now, bugs later	Readable and checkable team code
IEEE 754	fp32 for physics and measurement; bf16 only inside the network	fp16 overflows in 3D U-Nets; fp64 doubles memory	bf16 keeps the fp32 range
Open licences	LIDC-IDRI under CC BY 3.0 with citation [60]; PyTorch (BSD), crepes (BSD-3), Gradio (Apache 2.0)	Proprietary data or libraries: costly, restrictive	Legal reuse with attribution

Table 5.2: Hardware standards.

Standard	How it is applied	Alternatives	Rationale
NVIDIA CUDA platform	All GPU computation through PyTorch's CUDA backend	AMD ROCm: fewer cloud offers; CPU only: training would take weeks	Widest availability on rented GPUs
IEC 60601-1 / 60601-2-54 (X-ray equipment safety)	Applies to the DTS unit that would produce real data, not to our software	–	We rely on certified DTS hardware
ICRP 103 effective dose [61]	Doses compared in mSv for CT, low-dose CT and DTS	Air kerma: less meaningful for patients	Standard measure of patient risk

Table 5.3: Communication standards.

Standard	How it is applied	Alternatives	Rationale
HTTP/HTTPS with TLS	Application served locally over HTTP and to the browser through an HTTPS tunnel	Gradio share links (failed on Kaggle); SSH port forwarding (client setup)	Encrypted; works from any browser
JSON (RFC 8259)	Reports, results and calibration files	XML: verbose; pickle: unsafe to load	Human-readable, language-neutral
DICOM networking, HL7 FHIR	Future integration with hospital PACS and records	Custom APIs: no interoperability	Required in hospital workflows

5.2.2 Environmental Constraint

Energy and emissions. The L40S GPU has a 350 W power limit, so the 14-hour run used at most about 5 kWh on the GPU, roughly 7 kWh including CPU, memory and cooling, plus under 1 kWh for the CPU machine. At a world-average grid intensity of about 0.48 kg CO_2 per kWh (Bangladesh's grid is higher, about 0.6), the study emitted roughly 3.5–5 kg CO_2, comparable to a 20–30 km car trip. One measurement (about 45 s on the GPU) uses about 0.004 kWh. These are estimates from rated power, following [62], not metered values.

Data storage and waste. The 64 GB of downloaded DICOM was deleted after conversion to compact cases, and intermediate files that can be regenerated were not backed up. No hardware was bought, so the project produced no electronic waste.

Radiation as an environmental factor. Replacing CT follow-ups (about 1.5–6 mSv) with DTS (about 0.1 mSv) reduces the population's collective radiation dose.

Mitigations. A small model (17.4 M parameters) instead of a large generator; a prism instead of whole-volume generation; fixed compute budgets; CPU work on CPU machines; trained models reused for evaluation and the application instead of retraining; and, for future runs, cloud regions powered by renewable energy.

5.2.3 Ethical Constraint

Professional ethics. We followed the IEEE [63] and ACM [64] codes of ethics, in particular honesty about claims and results.

- **Honest claims:** all results come from *simulated* DTS, not real scanners, and Section 2.3 lists what we do not claim.
- **Honest results:** hypotheses were stated before the results, and the report gives whichever outcome we found, including a detector sensitivity of only 35% at two false positives per scan and the issues in Chapter 4.
- **Credit:** all datasets, libraries and ideas we build on are cited. **[FILL: keep or edit:]** AI coding assistants were used during implementation; every part of the code was reviewed, tested and verified by the team, which takes responsibility for it.
- **Fairness:** LIDC-IDRI comes mainly from US patients; performance on other populations, for example patients with healed tuberculosis scars, is unknown and is

stated as such.

- **Data ethics:** only de-identified public data, used under its licence, with no attempt at re-identification.

Professional responsibility. The team owns its decisions and mistakes. When the real-data pilot showed that the prism lost a third of its attenuation, we traced the cause to our own missing depth support and fixed it before the full run; when the first design of the project could not be verified, we narrowed it rather than present training losses as success. Each step's owner (Table 3.7) is accountable for its checks. The system is decision support: the responsibility for a diagnosis stays with the radiologist.

5.2.4 Health and Safety Constraint

Health impacts. *Physical:* the main benefit is lower radiation per follow-up (DTS about 0.1 mSv vs about 1.5 mSv for low-dose CT or several mSv for diagnostic CT), which matters because nodule follow-up repeats over years [6, 15]. The main physical risk is a wrong measurement that delays treatment of a growing cancer or leads to an unnecessary biopsy. *Mental:* living with an indeterminate nodule causes anxiety, and a falsely reassuring or falsely alarming number makes it worse.

Actionable solutions. Every number is reported with a calibrated interval and category probabilities so that the clinician sees when a result is uncertain; a low-confidence flag recommends CT; and results should be explained to patients by the clinician in categories rather than raw numbers. Table 5.4 lists the main hazards.

5.2.5 Social Constraint

In the short term, DTS-based follow-up could shorten CT waiting lists and reduce travel for patients outside major cities. In the long term it could make nodule surveillance, and later screening, feasible where CT capacity is the bottleneck, improving early diagnosis where the survival gain is largest. From the patient's perspective this means less dose and cost; from the radiologist's, a measurement tool with honest uncertainty; from the health system's, cheaper follow-up that needs training and local validation. Possible negative effects are over-diagnosis if many more people are followed, and a widening gap if only private hospitals adopt the tool; both argue for public-sector deployment and guideline-based use.

Table 5.4: Safety risk analysis (ISO 14971 style).

Hazard	Cause	Effect	Mitigation
Underestimated growth	Prior fills depth wrongly	Delayed treatment	Mass-first reporting; intervals; seen/unseen explanation; CT recommended when flagged
Overestimated growth	Vessel or chest wall included	Unnecessary CT or biopsy	Segmentation confined to a plausible extent; leaks reported as not measurable
Hallucinated nodule	Prior invents structure	False alarm	Data consistency with X-rays; clinician confirms
Missed nodule	Detector sensitivity 35% at 2 FP/scan	Nodule not suggested	Clinician can click anywhere; detector is an aid
Out-of-distribution use	Other population, scanner or dose	Silent errors	Validated setting stated; local validation and recalibration required
Over-trust	Automation bias	Clinician stops checking	Intervals and flags shown prominently; user training

5.2.6 Political Constraint

Adoption of health technology depends on public health policy and procurement. In Bangladesh, imaging equipment and software are purchased mainly through government tenders and large private hospital groups, so adoption would need evidence accepted by the health authorities and inclusion in national cancer-control plans. The use of foreign cloud computing for health data is itself a policy question, so a deployment should keep data and computation in the country.

5.2.7 Sustainability

Lifecycle plan. (1) Development and verification (this project); (2) retrospective validation on real DTS with CT ground truth from a partner hospital; (3) a prospective reader study; (4) regulatory registration; (5) deployment with monitoring of measurement errors and interval coverage; and (6) periodic recalibration of the conformal layer, which needs only new calibration cases, not retraining, when scanners or populations change.

Three dimensions. *Environmental:* small model, low energy per measurement, no special hardware. *Economic:* runs on one mid-range GPU or a low-cost cloud instance, and open-source code avoids licence fees. *Social:* documented and tested code that another team can maintain, and a method that extends to other limited-angle modalities.

5.2.8 Legal Considerations

- **Data licence:** LIDC-IDRI is distributed by TCIA under the Creative Commons Attribution 3.0 licence and the TCIA data usage policy [60]; we cite the required publication [19] and make no attempt at re-identification.
- **Medical device regulation:** software that measures nodules for diagnosis is software as a medical device. In Bangladesh, medical devices are registered by the Directorate General of Drug Administration (DGDA) **[FILL: verify the current DGDA rules for software]**. Elsewhere the US FDA (510(k) or De Novo), the EU Medical Device Regulation [65] and the EU AI Act [66] would apply; each requires risk management [58], life-cycle processes [57] and clinical evidence.
- **Data protection:** a clinical deployment processes personal health data and must follow Bangladesh's personal data protection law **[FILL: verify the current act and its status]** and, for international data, GDPR and HIPAA: consent, purpose limitation and secure storage. The demo uses no personal data.
- **Liability:** the intended use is decision support; the clinician remains responsible for the diagnosis.

5.2.9 Cultural Considerations

- **Language and literacy:** a deployed interface should offer Bangla and present results in plain categories, with numbers available for specialists.
- **Family-centred decisions:** diagnoses are often discussed with the family, so uncertainty should be explained through the clinician in a way the family can follow, not as raw probabilities.
- **Stigma:** lung cancer is often linked to smoking and blame; communication should avoid blame, and many patients, especially women exposed to cooking smoke, have never smoked.
- **Local disease patterns:** healed tuberculosis leaves calcified granulomas that resemble nodules. The model was not trained on such patients, so local validation is required before use, which also respects patients' right to a tool tested on people like them.
- **Trust:** many patients trust the doctor more than a machine; keeping the clinician in charge respects this.

5.3 Cost Analysis

Actual project budget. Table 5.5 gives the actual cost of FYDP-III (exchange rate assumed: US$1 ≈ BDT 120).

Table 5.5: Actual project budget.

Item	US$	BDT
RunPod GPU (L40S, 14-hour full run incl. network volume)	≈ 16	≈ 1,900
RunPod CPU machine (download and case building)	< 2	< 240
Kaggle notebooks (pilot, application)	0	0
LIDC-IDRI data and open-source software	0	0
Total	**about 18**	**about 2,150**

Alternate budget: buy instead of rent. A workstation with one RTX 4090 (24 GB) would cost about BDT 4.5–5.5 lakh **[FILL: check the current local price]**. It would run the study in 20–30 hours but pays off only after hundreds of runs. Renting cost less than 1% of buying and provided a larger 48 GB GPU, so renting was the right choice for a one-off study; buying makes sense only for a centre with continuous use.

Revenue model (illustrative). For a commercial version, assume 18 months of validation and registration and customers among diagnostic centres and hospitals with DTS-capable radiography (Tables 5.6 and 5.7).

Table 5.6: Illustrative revenue (BDT lakh).

Revenue stream (price)	Year 1	Year 2	Year 3
Pay per study (BDT 300 per nodule follow-up)	0	18 (6,000 studies)	60 (20,000 studies)
Hospital subscription (BDT 6 lakh per year)	0	18 (3 hospitals)	48 (8 hospitals)
Total revenue	**0**	**36**	**108**

Table 5.7: Illustrative expenses and cash flow (BDT lakh).

Item	Year 1	Year 2	Year 3
Development team (3 engineers)	25	25	30
Clinical validation study	20	10	–
Regulatory registration and quality system	8	4	4
Cloud / GPU and hosting	3	5	8
Marketing and training	2	6	8
Total expenses	**58**	**50**	**50**
Net cash flow	−58	−14	+58
Cumulative cash flow	**−58**	**−72**	**−14**

Under these assumptions the venture breaks even early in year four. The largest uncertainty is the cost and duration of clinical validation, which is why a partnership with a public hospital is the first step.

5.4 Complex Engineering Problem

MedLift-3D is a complex engineering problem: it is ill-posed by physics, combines X-ray physics, generative AI and statistics, has conflicting requirements (dose, accuracy, honesty, cost), and has direct consequences for patients.

5.4.1 Complex Problem Solving

Table 5.8: Mapping with complex problem solving.

P1 Depth of knowledge	P2 Range of conflicting requirements	P3 Depth of analysis	P4 Familiarity of issues	P5 Extent of applicable codes	P6 Stakeholder involvement	P7 Inter-dependence
✓	✓	✓	✓	✓	○	✓

P1 – Depth of knowledge. The solution requires X-ray projection physics, linear algebra of range and null spaces, probabilistic diffusion models, conformal statistics and 3D deep learning, beyond the standard curriculum. Table 5.9 maps the knowledge profiles.

Table 5.9: Knowledge profile mapping for P1.

Knowledge profile	Where it is used
WK1 Natural sciences	X-ray attenuation, Poisson photon statistics, radiation dose
WK2 Mathematics	Projection moments and conditioning; CGLS; probability and exchangeability for conformal prediction
WK3 Engineering fundamentals	Numerical methods (SIRT, conjugate gradients), algorithms, data structures
WK4 Specialist knowledge	Tomography, diffusion models, 3D U-Nets, Gaussian representations
WK5 Engineering design	Requirements, prism design, verification steps, application
WK6 Engineering practice	Cloud GPU operation, testing, version control, reproducibility
WK7 Engineering in society	Radiation safety, regulation, ethics, cultural factors
WK8 Research literature	Literature on DTS, sparse-view CT, diffusion, Gaussians and conformal prediction

P2 – Range of conflicting requirements. Dose vs accuracy, filling the missing wedge vs never inventing anatomy, single numbers vs honest intervals, memory vs field of view, and detection recall vs false positives (Table 3.2).

P3 – Depth of analysis. We derived from first principles which quantities DTS can measure, built a 583-patient virtual trial, and analysed calibration, size-group coverage and seen/unseen content rather than image quality alone.

P4 – Familiarity of issues. No standard solution exists for calibrated 3D nodule measurement from DTS; the combination of a data-consistent prior inside a DTS prism, Gaussian mass and conformal calibration had to be designed for this problem.

P5 – Extent of applicable codes. DICOM, IEC 62304, ISO 14971, data licences and medical device regulation (Table 5.1).

P6 – Extent of stakeholder involvement (partial). Clinical needs were taken from guidelines and supervisor feedback; direct evaluation by radiologists is future work.

P7 – Interdependence. Simulation, reconstruction, prior, Gaussians, calibration and detector depend on each other: the depth-support bug in reconstruction, for example, corrupted the training data and every downstream measurement until it was fixed.

5.4.2 Engineering Activities

Table 5.10: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓	✓	✓	✓	✓

A1 – Range of resources. A public archive of 583 patients (about 64 GB), rented CPU and GPU machines with shared storage, free Kaggle GPUs and open-source libraries, all within about US$18.

A2 – Level of interaction. Interaction among three team members with distinct roles, the supervisor, the course teacher, published clinical guidelines and open-source communities (crepes, Gradio, pylidc).

A3 – Innovation. The mass-first projection-moment argument for DTS, the prism-based data-consistent diffusion prior with Gaussian mass, decision-level calibrated uncertainty, and a truth-referenced seen/unseen test.

A4 – Consequences for society and environment. Lower radiation and cost per follow-up and wider access to nodule surveillance, balanced against the risk of wrong measurements, which intervals, flags and clinician control address (Table 5.4).

A5 – Familiarity. Diffusion models, conformal predictive systems and null-space analysis go well beyond typical undergraduate projects and were learned during the project.

5.5 Summary

MedLift-3D follows the software, hardware and communication standards relevant to medical imaging software and prepares for IEC 62304 and ISO 14971. Its main impacts are positive (lower dose and cost, wider access), with clear risks (wrong measurements, over-trust, untested populations) that the design mitigates through calibrated intervals, flags, clinician control and a validation plan. The project cost about US$18 and meets the attributes of a complex engineering problem.

Chapter 6

Conclusion

This chapter summarises what MedLift-3D achieved, states its limitations and future work, and reflects on the independent learning the project required.

6.1 Summary

MedLift-3D turns one click on a low-dose chest tomosynthesis image into a 3D nodule measurement with calibrated uncertainty. We showed from projection moments that DTS measures a nodule's mass in every view but its depth extent about 15 times worse, and built the system around that fact: a prism reconstruction around the click, a 17.4-million-parameter conditional diffusion prior kept consistent with the X-rays, 3D Gaussians that give mass in closed form, conformal intervals and calibrated size-category probabilities, a learned detector to suggest clicks, and a seen/unseen explanation of every result. A verified 18-step pipeline ran a 583-patient virtual clinical trial on LIDC-IDRI in 14 hours for about US$16. The prior improved nodule-region PSNR from 11.88 to 15.23 dB without overfitting, the detector reached a CPM of 0.298 against 0.017 for a training-free baseline, and every requirement with a fixed criterion passed its check. **[FILL: one sentence with the H1–H6 outcomes]**

6.2 Limitation

- **Simulation only.** No public chest DTS dataset with CT ground truth exists; our DTS is simulated from CT with a monochromatic beam and no scatter, so performance on real scanners is unknown.
- **Acquisition.** 15 views instead of the 60–74 of clinical units (at equal dose) and a conservative 30° sweep; the prior was trained and tested on this geometry only.

- **Detector sensitivity.** 35% at two false positives per scan is too low for unaided use; the detector is an aid only.
- **Growth.** Virtual growth scales the nodule uniformly; real nodules grow irregularly.
- **Ground truth.** LIDC labels vary between radiologists (consensus volume / mean reader volume 0.81), and only nodules of 4–30 mm marked by at least three readers were used.
- **Population.** LIDC is mostly from US patients; populations with tuberculosis scarring were not represented.
- **Training budget.** The prior trained for 6 hours and its held-out loss was still falling, so it is probably under-trained.
- **Patient thickness** was taken from the CT body outline; a real DTS unit would have to measure it.

6.3 Future Work

1. **Real DTS data:** partner with a hospital that has a DTS unit to collect paired DTS and CT scans, recalibrate the conformal layer on them, and measure real performance.
2. **Physics realism:** add polychromatic spectra, scatter and the 60–74-view clinical protocols to the simulator.
3. **Longer and larger training:** train the prior longer with augmentation, test larger models, and train the detector with more nodule-free negatives.
4. **Real growth:** use longitudinal CT pairs instead of virtual growth.
5. **Reader study:** measure whether radiologists make better follow-up decisions with MedLift-3D than with DTS alone.
6. **Deployment:** a Bangla interface, DICOM/PACS integration, an on-site GPU option, and a quality system towards regulatory registration.
7. **Other limited-angle problems:** breast tomosynthesis and dental cone-beam CT share the missing-wedge structure.

6.4 Independent Learning and Reflection

Curiosity. The project began as a generative-modelling idea, but we kept asking a deeper question: what can a 30° X-ray sweep actually measure? Following that question

into projection moments led to the mass-first design. The same curiosity made us ask whether the prior recovers or invents depth, which became the seen/unseen analysis, and when we found that raw ensemble spread is not a probability we studied conformal prediction until we could produce calibrated category probabilities.

Self-initiative. Beyond the required scope we added a learned detector after the training-free one mostly found vessels, a fit check to show that the prior does not overfit, conformal predictive systems with size-group calibration, explainability with a truth-referenced test, a geometry-sensitivity study to justify the 15-view choice, and an automatic HTML report so that every check is visible to examiners.

Independence. Outside the classroom we learned to run a 14-hour unattended job on rented cloud machines (CPU and GPU pods sharing a network volume, terminal multiplexing, resumable scripts and backups), to download and organise 64 GB of medical images from TCIA, to read radiologist annotations with pylidc, and to serve a GPU application from a Kaggle notebook through an HTTPS tunnel when the usual share link failed.

Transfer of past learning. Table 6.1 shows how earlier coursework was applied.

Table 6.1: Earlier coursework applied in MedLift-3D.

Earlier course knowledge	Application in MedLift-3D
Linear algebra (range, null space, least squares)	Seen/unseen split with CGLS; SIRT as an iterative least-squares solver
Calculus and moments	Projection-moment argument: mass is the 0th moment, depth extent the 2nd
Probability and statistics	Poisson noise, conformal prediction, Brier score, rank correlations, bootstrap confidence intervals
Numerical methods	Iterative solvers, numerical stability, mixed precision
Data structures and algorithms	Ray marching, candidate ranking, patient-level splitting
Artificial intelligence and machine learning	Diffusion training, U-Nets, overfitting checks, train/calibration/test discipline
Image processing and computer graphics	Projection geometry, 3D Gaussians, segmentation, saliency maps
Software engineering	Requirements and traceability, unit tests, version control, the verified pipeline
Computer networks	HTTPS tunnels and port forwarding for the application

Reflection on learning. Our main lesson was that a large architecture with falling training losses does not show that a single clinical measurement is right. Rebuilding

the work around measurable clinical numbers, explicit hypotheses and automatic checks changed how we think about engineering: a result counts only if someone else can verify it, and a negative result reported honestly is worth more than a claim that cannot be tested. We also learned to work within real limits of time, money and data, to plan computation like a budget, and to treat uncertainty as part of the answer rather than an embarrassment. These habits, more than any single model, are what we take from the project into our careers.

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